Simulation results

This empirical-resampling evaluation uses the symmetrized Jöreskog BMI twin tables as the data-generating distributions. It includes 50, 100, and 500 pairs per zygosity group. The stored output, `simulation/sim_ace_joreskog-2.rds`, reports 2,400 analyzed replications per condition. The design is an empirical-resampling evaluation, not a simulation from known ACE parameters: the reference values are two-step polychoric estimates from the symmetrized tables, and the generating distributions need not satisfy the fitted Gaussian liability model with common thresholds.

The methods are Stan with simplex ACE proportions (`stan`), Stan with a regularizing simplex prior (`stan_reg`), Stan with unrestricted twin correlations (`stan_nc`), and UMX/OpenMx (`umx`). All three Stan models use standard-normal threshold priors; their ACE/correlation priors and parameter spaces differ. In Stan, convergence requires no divergent transitions, $\hat{R} < 1.01$, and bulk effective sample size greater than 400 for the reported ACE quantities. Stan retries with increasing `adapt_delta` and `max_treedepth`. UMX convergence records successful optimizer status and finite estimates; standard-error availability, boundary estimates, and profile-interval success are distinct outcomes. A zero variance component is an admissible boundary estimate. Point-estimate metrics use accepted fits; interval coverage and width use fits with available intervals. The reported interval counts make those denominators explicit.

In the tables, **Simplex**, **Regularized simplex**, and **Unrestricted** denote the three Stan methods above, and **UMX** denotes UMX/OpenMx. All reported intervals have nominal 90% coverage. Stan point estimates are posterior means.

Results overview

Convergence was high across the evaluated conditions, ranging from 95.0% to 100.0% (Table 1). Both simplex Stan methods and UMX produced admissible estimates in all accepted fits. The unrestricted-correlation model produced admissible posterior-mean estimates in 57.3%, 63.7%, and 84.4% of accepted fits at 50, 100, 500 pairs per group, respectively. UMX estimates remained admissible when a variance component reached zero: the c^2 boundary occurred in 37.8%, 32.7%, and 14.6% of accepted UMX fits. These boundary estimates should be distinguished from optimizer failure and inadmissibility.

UMX standard errors for both components were available in at least 99.8% of attempts, but successful profile-interval computation was much less frequent (Table 2). Both component intervals succeeded in 0.6%, 6.8%, and 26.5% of attempts as sample size increased. At 50 pairs per group, h^2 and c^2 profile intervals were available in 83 and 32 of the 2,400 attempts, respectively. Consequently, UMX coverage and mean-width results describe the subset with successful interval computation, rather than all accepted fits.

The estimation results illustrate tradeoffs between bias, variability, and admissibility (Table 3; Table 4). At 50 pairs per group, the unrestricted model had smaller absolute biases for both

components than the simplex methods, but larger RMSEs and frequent inadmissible estimates. The regularized simplex method had more negative h^2 bias and greater h^2 RMSE than the basic simplex method across the evaluated sample sizes; its h^2 interval coverage was also lower. Bias magnitudes and RMSEs generally decreased with sample size. These findings characterize this empirical-resampling evaluation: the reference components are descriptive targets, and the results do not establish a general ranking under a correctly specified ACE model.

Table 1: Convergence, admissibility, and median analysis time. Each method has 2,400 attempted replications per condition. Convergence uses all attempts; admissibility uses accepted fits.

Method	Accepted	Convergence (%)	Admissible (%)	Median time (s)
50 pairs/group				
Simplex	2281	95.0	100.0	1.69
Regularized simplex	2374	98.9	100.0	1.59
Unrestricted	2334	97.2	57.3	1.36
UMX	2393	99.7	100.0	0.63
100 pairs/group				
Simplex	2400	100.0	100.0	1.76
Regularized simplex	2400	100.0	100.0	1.70
Unrestricted	2400	100.0	63.7	1.41
UMX	2399	100.0	100.0	0.79
500 pairs/group				
Simplex	2400	100.0	100.0	1.89
Regularized simplex	2400	100.0	100.0	1.74
Unrestricted	2400	100.0	84.4	1.37
UMX	2400	100.0	100.0	0.54

Median time is per attempted analysis, including retries and interval computations when requested, and excludes compilation and data generation.

Table 2: UMX boundary and interval outcomes. The c^2 boundary rate uses accepted fits; SE and joint profile-CI success rates use all 2,400 attempts per condition.

Pairs/group	c^2 boundary (%)	SE success (%)	Both profile CIs succeed (%)
50	37.8	99.8	0.6
100	32.7	100.0	6.8
500	14.6	100.0	26.5

UMX SE success requires standard errors for both components, and joint profile-CI success requires both component profile intervals to succeed. The component-specific interval counts are shown in Table 3 and Table 4. Bias and RMSE use accepted fits, including inadmissible estimates from the unrestricted model. Coverage and mean interval width are conditional on successful interval computation, so UMX coverage should be interpreted alongside its interval counts. Each CI count is out of 2,400 attempted replications.

Table 3: Estimation performance for h^2. Coverage and mean width use available 90% equal-tailed posterior intervals (Stan) or profile-likelihood intervals (UMX). CI count is the number of available intervals out of 2,400 attempts.

Method	Bias	RMSE	Mean CI width	Coverage (%)	CI count
50 pairs/group					
Simplex	−0.223	0.253	0.601	78.1	2281
Regularized simplex	−0.259	0.279	0.555	57.1	2374
Unrestricted	−0.014	0.322	1.104	90.5	2334
UMX	−0.083	0.238	0.754	97.6	83
100 pairs/group					
Simplex	−0.153	0.194	0.488	83.9	2400
Regularized simplex	−0.193	0.221	0.470	68.0	2400
Unrestricted	−0.004	0.236	0.768	89.4	2400
UMX	−0.052	0.182	0.637	86.5	342
500 pairs/group					
Simplex	−0.047	0.084	0.260	92.4	2400
Regularized simplex	−0.070	0.097	0.257	86.8	2400
Unrestricted	−0.006	0.103	0.339	90.0	2400
UMX	−0.014	0.092	0.302	88.8	1388

Table 4: Estimation performance for c^2. Coverage and mean width use available 90% equal-tailed posterior intervals (Stan) or profile-likelihood intervals (UMX). CI count is the number of available intervals out of 2,400 attempts.

Method	Bias	RMSE	Mean CI width	Coverage (%)	CI count
50 pairs/group					
Simplex	0.172	0.198	0.499	92.1	2281
Regularized simplex	0.193	0.211	0.465	81.0	2374
Unrestricted	−0.043	0.292	0.982	90.7	2334
UMX	0.075	0.208	0.705	78.1	32
100 pairs/group					
Simplex	0.127	0.160	0.414	91.4	2400
Regularized simplex	0.155	0.177	0.400	82.5	2400
Unrestricted	−0.022	0.215	0.698	89.9	2400
UMX	0.048	0.162	0.565	49.3	221
500 pairs/group					
Simplex	0.042	0.073	0.234	94.2	2400
Regularized simplex	0.062	0.084	0.231	88.8	2400
Unrestricted	0.001	0.094	0.314	90.9	2400
UMX	0.014	0.083	0.285	68.1	636

Joint estimates and admissibility

Figure 1, Figure 2, and Figure 3 show the joint distributions of the component estimates among accepted fits. The axes are shared across methods and sample sizes and include all plotted estimates. The shaded triangle marks the admissible region: $h^2 \geq 0$, $c^2 \geq 0$, and $h^2 + c^2 \leq 1$. Dashed lines mark the empirical-reference values, rather than known generating ACE parameters. UMX zero-boundary estimates are admissible; red triangles identify inadmissible estimates from the unrestricted-correlation model.

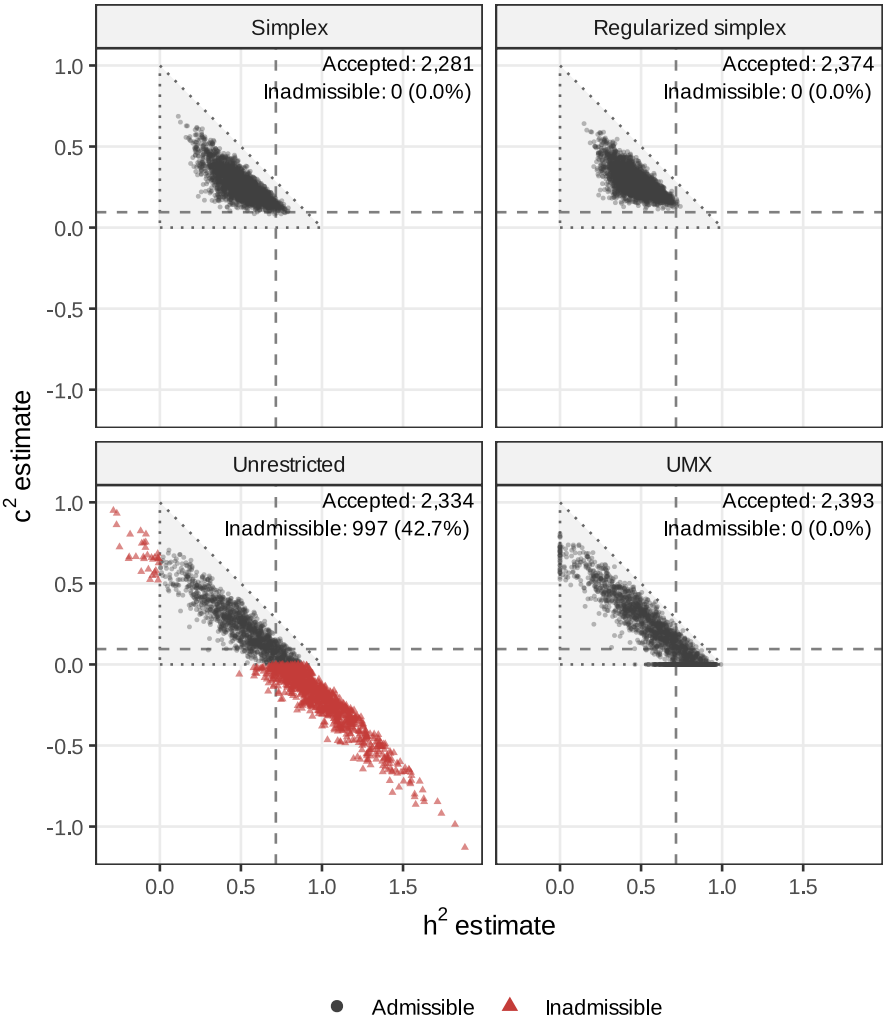


Figure 1: Joint h^2 and c^2 estimates with 50 pairs per zygosity group, among accepted fits. Stan estimates are posterior means; UMX estimates are standardized variance proportions. The dotted triangle bounds the admissible region, red triangles mark inadmissible estimates, and dashed lines mark the empirical-reference values. Percentages in each panel use accepted fits.

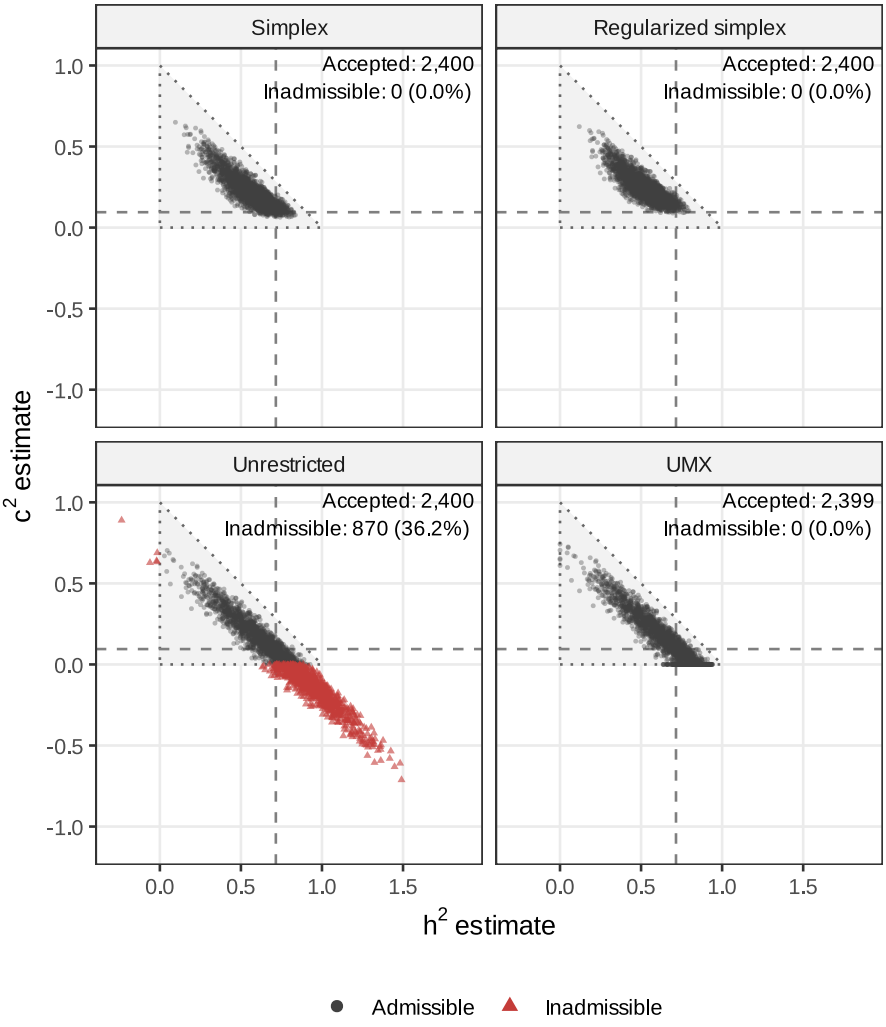


Figure 2: Joint h^2 and c^2 estimates with 100 pairs per zygosity group, among accepted fits. Point summaries, boundaries, reference lines, and percentage denominators are as in Figure 1.

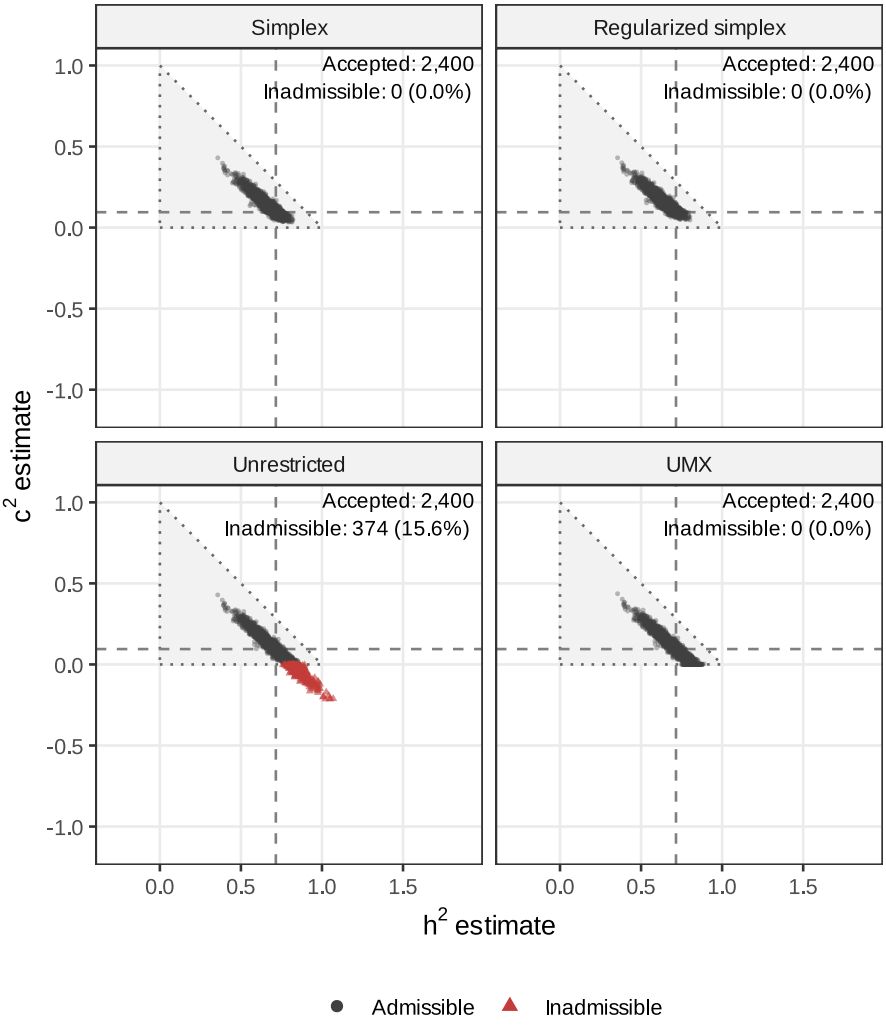


Figure 3: Joint h^2 and c^2 estimates with 500 pairs per zygosity group, among accepted fits. Point summaries, boundaries, reference lines, and percentage denominators are as in Figure 1.